Using Machine Learning to Predict and Analyze Cyclone Events: A Data-Driven Approach to Tropical Storm Forecasting

MSc DS CST4090

Gautam Kumar Rai Ramessur
Dr Zia Lallmahomed (Supervisor)

Middlesex University

INTRODUCTION

Tropical cyclones are among the most destructive natural hazards, causing major socio-economic disruption, displacement, and loss of life (Emanuel, 2017). Despite advances in numerical weather prediction (NWP), forecasting cyclone intensity and rapid changes remains challenging, especially for vulnerable coastal and island regions (IPCC, 2021).

Machine learning offers new opportunities by capturing nonlinear relationships between atmospheric and oceanic variables, overcoming some limitations of traditional models. Deep learning approaches further enable the integration of sequential storm data, improving accuracy and timeliness of forecasts.
(References: Emanuel, 2017; IPCC, 2021)

Objectives

- Collect, preprocess, and integrate cyclone datasets (IBTrACS & ERA5).
- Develop ML & DL models: RF, XGB, LightGBM, SVM, LSTM, TCN.
- Test hypotheses:
- ▸ Storm duration vs intensity
- ▸ Latitude vs intensity
- ▸ Pressure vs intensity
- Evaluate performance using RMSE, MAE, R^2.

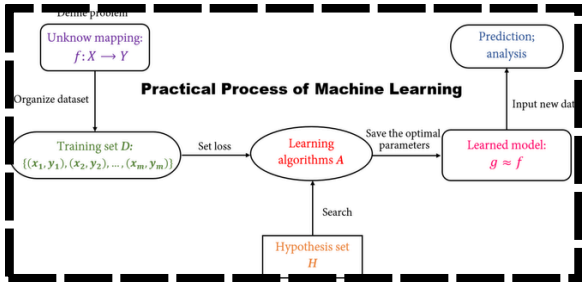

Fig. 1: Methodological Workflow — Machine Learning for Cyclone Forecasting:

Methodology (Flow Diagram)

- Data Acquisition: Tropical cyclone tracks from IBTrACS were integrated with environmental variables (SST, MSLP, wind) from ERA5.
- Preprocessing: Datasets were cleaned, interpolated, aligned temporally and spatially, and normalised for consistency.
- Feature Engineering: Derived features included storm duration, average latitude, SST anomalies, vertical wind shear, and lagged intensity.
- Hypothesis Testing: Relationships between predictors (duration, latitude, pressure) and cyclone intensity were assessed using Pearson and Spearman correlations.
- Model Training: Classical ML (Random Forest, XGBoost, LightGBM, SVR) and DL models (LSTM, TCN) were implemented in Python/Google Colab.
- Evaluation: Models were benchmarked with RMSE, MAE, R^2, and interpretability assessed via SHAP values.

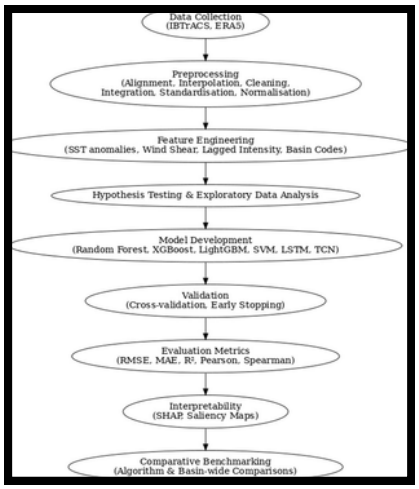

Fig 2 : Methodological flowchart outlining data acquisition, preprocessing, hypothesis testing, model development, and evaluation stages.

Exploratory Data Analysis (Graphs)

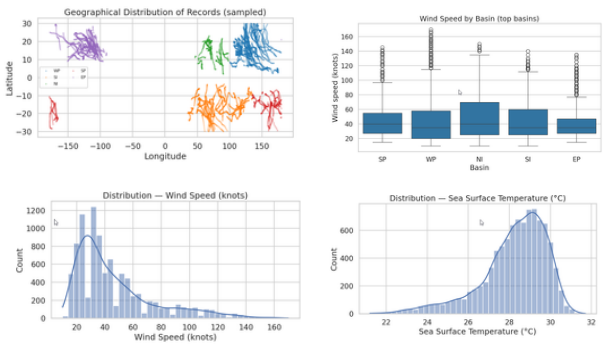

Fig. 3:Graph for EDA

Hypothesis Testing

- H1: Duration vs Intensity → moderate positive correlation.
- H2: Latitude vs Intensity → weak/insignificant.
- H3: Pressure vs Intensity → very strong inverse relationship.

Hypothesis	Scientific Rationale	Pearson (r, p)	Spearman (ρ, p)	Result	Interpretation
H1: Cyclones that last longer reach higher intensity	Longer duration allows storms to interact with favourable SST, low shear, and moisture → greater intensification	r = 0.51, p < 0.001	ρ = 0.62, p < 0.001	✅ Accepted	Moderate–strong positive association. Duration is a **secondary but useful predictor** of maximum wind speed.
H2: Cyclones near the equator tend to be stronger	Latitude influences cyclogenesis via Coriolis effect, but post-formation role uncertain	r = −0.10, p = 0.157	ρ = −0.10, p = 0.139	❌ Rejected	Weak, insignificant correlation. Latitude matters for **formation**, not for peak intensity after formation.
H3: Lower central pressure leads to higher cyclone intensity	Pressure–wind relationship is fundamental to cyclone dynamics	r = −0.97, p < 0.001	ρ = −0.96, p < 0.001	✅ Strongly Accepted	Near-perfect inverse correlation. **Primary predictor** of cyclone strength; validates forecasting practice.

Fig. 4: Hypothesis Results

Results (Model Comparison)

Table 4.1: Model Performance Summary

Model	RMSE	MAE	R^2
Random Forest	9.84	6.36	0.878
XGBoost	8.45	5.64	0.910
LightGBM	7.72	5.23	0.925
SVR	12.78	7.71	0.794
LSTM	15.83	11.02	0.673
TCN (Run 1)	20.34	13.38	0.461
TCN (Run 2)	18.75	12.77	0.542

Fig. 5: Model Performance Summary

CONCLUSIONS

This study demonstrates that machine learning can significantly enhance cyclone forecasting by capturing nonlinear relationships between storm attributes and environmental drivers. Among predictors, minimum central pressure emerged as the strongest determinant of cyclone intensity, consistent with established meteorological theory (Knutson et al., 2020). Storm duration showed a secondary but meaningful role, while latitude proved weak, aligning with previous findings that latitude influences cyclone genesis rather than intensity (Emanuel, 2017). Boosting algorithms (LightGBM, XGBoost) consistently outperformed deep learning models, reflecting their ability to prioritise structural features such as pressure and storm longevity. These results highlight the operational value of ML-driven approaches in strengthening early warning systems and supporting climate resilience, especially for vulnerable small island and coastal regions (IPCC, 2021).

REFERENCES

- *Emanuel, K. (2017). Will global warming make hurricane forecasting more difficult? Bulletin of the American Meteorological Society, 98(3), 495–501.*
- *Hersbach, H., Bell, B., Berrisford, P., et al. (2020). The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society, 146(730), 1999–2049.*
- *IPCC. (2021). Climate Change 2021: The Physical Science Basis. Cambridge University Press.*
- *Knapp, K. R., Kruk, M. C., Levinson, D. H., et al. (2010). The International Best Track Archive for Climate Stewardship (IBTrACS). Bulletin of the American Meteorological Society, 91(3), 363–376.*
- *Knutson, T., Camargo, S. J., Chan, J. C. L., et al. (2020). Tropical cyclones and climate change assessment. Bulletin of the American Meteorological Society, 101(3), E303–E332.*